

Semi-Supervised Learning with Correlated Convolutional Networks

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Abstract

One of the celebrated successes of supervised deep learning has been the ubiquitous application of convolutional neural networks (CNNs) to computer vision tasks, most notably large-scale image classification where many labeled samples are available. When few labels are available, *semi-supervised* learning aims to leverage large amounts of unlabeled samples in order to improve predictive models.

We propose Correlated Convolutional Networks (CCN) which is based on recent multi-view approaches to semi-supervised learning. CCN has two distinct training modalities, an unsupervised mode based on deep CCA and a supervised mode which uses the extracted canonical correlations as a regularizer. In the shallow setting, this procedure has been proven to reduce the variance of an estimator while incurring little additional bias. We evaluate CCN on a variety of image classification tasks. Compared with a baseline CNN with as many parameters, appropriately trained using batch-normalization and regularized using dropout, CCN obtains higher accuracy faster and with lower variance.

1. Introduction

Deep learning has achieved remarkable success in many different problem domains LeCun et al. (2015). However, one well-known limitation of such models as they get larger and more expressive is the increase in amount of data required to effectively train them. In tasks like natural image classification, modern neural networks use thousands of training samples per class. Obtaining this amount of labeled data is costly and time-consuming and may not be possible for all problem domains. For example medical image analysis is often limited to a handful of samples, as it involves laboratory experiments and hours of expert analysis to obtain labeled data.

Semi-supervised learning aims to learn good predictive models by effectively using the potentially large amounts of *unlabeled* data together with relatively few *labeled* samples for supervision. A recent research direction of particular interest tackles semi-supervised learning as a combination of classical unsupervised and supervised techniques. McWilliams et al. (2013) proposed Correlated Nyström Views (XNV), which is based on theoretically motivated approaches to semi-supervised learning such as co-training Mitchell and Blum (1998) and multi-view learning Kakade and Foster (2007). These works show that under relatively weak conditions, given two *views* of the data, it is possible to learn a provably good basis for supervised learning using canonical correlation analysis (CCA) – a purely unsupervised technique – by finding the directions of largest correlation between these views. In particular, XNV uses randomized kernel approximations to generate good views of the data and leverages the theoretical guarantees of multi-view learning and translates them into a fast, practical semi-supervised algorithm. However, using randomized kernel approximations as in XNV, may not yield the best possible feature representations for complex tasks such as image classification or language modelling. For these tasks deep models are preferred as they learn better representations of the data.

Building on recent approaches to deep CCA (Andrew et al., 2013), we propose Canonically Correlated Networks (CCN) a deep extension of XNV. CCN instead uses neural networks to generate two adaptive views of the underlying data. We introduce the *canonical softmax* layer which imposes a penalty on weights which have low correlation between the learned input views. This results in a semi-supervised algorithm that allows to efficiently train good view representations with just a small fraction of the labeled data used in a fully-supervised setting.

Outline. In the following section we review multi-view semi-supervised learning using CCA. In Section 3 we detail our proposed CCN network architecture and the specifics of how both the unsupervised and supervised objectives are trained. We show results of CCN applied to various image classification problems in Section 4. We observe that CCN can improve sample efficiency and generalisation error compared with a baseline model of similar size when the number of labeled points is small.

We discuss our contribution in the context of recent deep learning approaches to semi-supervised learning in Section 5. Finally, we conclude with a discussion of the benefits and weaknesses of the CCN architecture and highlight some interesting avenues for future investigation.

2. Learning with Multiple Views

Suppose we have a dataset $\mathcal{D} = \{\mathcal{D}_L, \mathcal{D}_U\}$ consisting of a *labeled* portion $\mathcal{D}_L = \{\mathbf{x}_i, y_i\}_{i=1}^n$ and a much larger *unlabeled* portion $\mathcal{D}_U = \{\mathbf{x}_i\}_{i=n+1}^N$ with $\mathbf{x}_i \in \mathbb{R}^D$, $y_i \in \mathbb{R}$. Multi-view learning is concerned with the setting where there are two different *views* on the data. Abstractly we define a view as a function which maps the data to a hypothesis space $\mathcal{H}$,

$$\mathbf{z}^{(\nu)} : \mathbb{R}^D \rightarrow \mathcal{H}^{(\nu)} = \mathbb{R}^d : \mathbf{x} \mapsto \mathbf{z}^{(\nu)}(\mathbf{x}) =: \mathbf{z}^{(\nu)} \quad \text{for } \nu \in \{1, 2\}.$$

For example, if $\mathcal{D}$ is a corpus of news articles, $\mathbf{z}^{(1)}$ might correspond to the images and $\mathbf{z}^{(2)}$ the accompanying text for a given topic.

2.1 Canonical Correlation Analysis

Canonical Correlation Analysis (CCA) (Hotelling, 1936) is an unsupervised method to discover a data subspace which maximizes the correlation between two sets of variables. It can be seen as an extension of principal components analysis (PCA) to two sets of random variables instead of one: CCA extracts directions of maximum correlation between the two views. Recently, CCA has seen renewed interest, in particular in language modelling, word embeddings (Dhillon et al., 2012) and applications in bioinformatics (Witten et al., 2009).

Given two matrices¹ $Z_\nu \in \mathbb{R}^{n \times d_\nu}$ whose rows are $\mathbf{z}^{(\nu)}(\mathbf{x}_i)^\top$ for $\nu = \{1, 2\}$, CCA finds pairs of linear projection vectors $\mathbf{w}^{(1)}$ and $\mathbf{w}^{(2)}$, which maximize the correlation of the projected views. The optimization problem to obtain the first pair of CCA weights is:

$$(\mathbf{w}^{(1)*}, \mathbf{w}^{(2)*}) = \underset{\mathbf{w}^{(1)}, \mathbf{w}^{(2)}}{\operatorname{argmax}} \operatorname{corr}(Z_1 \mathbf{w}^{(1)}, Z_2 \mathbf{w}^{(2)}). \quad (1)$$

Subsequent pairs forming the *canonical bases* $W_1 = [\mathbf{w}_1^{(1)}, \dots, \mathbf{w}_d^{(1)}]$ and $W_2 = [\mathbf{w}_1^{(2)}, \dots, \mathbf{w}_d^{(2)}]$ are obtained by first defining the the cross- and auto-covariance matrices of the (mean-centered) views: $\Sigma_{12} = Z_1^\top Z_2$, $\Sigma_{11} = Z_1^\top Z_1$, $\Sigma_{22} = Z_2^\top Z_2$, and reformulating the problem using the trace norm objective:

$$(W_1^*, W_2^*) = \underset{W_1, W_2}{\operatorname{argmax}} \operatorname{tr}(W_1^\top \Sigma_{12} W_2) \quad \text{s.t. } W_1^\top \Sigma_{11} W_1 = W_2^\top \Sigma_{22} W_2 = I. \quad (2)$$

The constraint ensures that subsequent projection vectors are orthogonal. The solution to (2) is obtained by the singular value decomposition (SVD) of $\Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1/2} = W_1 \Lambda W_2^\top$ where $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_d)$ is the matrix of canonical correlations (Knutsson et al., 1998).

Deep approaches to nonlinear CCA have also been proposed (Andrew et al., 2013; Wang et al., 2016), which learn nonlinear, correlation maximizing transformations of each view as a set of parameters of an autoencoder model. We will briefly review these in Section 3 when we discuss our contribution.

1. To simplify notation we will assume $d_1 = d_2 = d$.

2.2 Semi-supervised multi-view learning

Given two prediction functions $f^{(1)}$ and $f^{(2)}$ which produce hypotheses based on $\mathbf{z}^{(1)}$ and $\mathbf{z}^{(2)}$ respectively, multi-view learning works by optimising a supervised loss on $\mathcal{D}_L$ while constraining $f^{(1)}(\mathbf{z}^{(1)})$ to be similar to $f^{(2)}(\mathbf{z}^{(2)})$ on unlabeled samples, $\mathcal{D}_U$. Intuitively, this reduces the complexity of the supervised learning problem by penalizing solutions where mismatches between the views occur. *Canonical ridge regression* achieves this by imposing a penalty on the *CCA norm*.

Canonical ridge regression first estimates the canonical basis using a large amount of unlabeled data then using labeled data finds regression coefficients $\hat{\beta}^{(\nu)} \in \mathbb{R}^d$ in the canonical basis subject to a constraint on the CCA norm

$$\hat{\beta}^{(\nu)} = \underset{\beta}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n (y_i - \beta^\top (W_\nu^\top \mathbf{z}_i^{(\nu)}))^2 + \|\beta\|_{\text{CCA}}^2. \quad (3)$$

Where the CCA norm has the following definition.

Definition 1 (CCA norm) Let $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_d)$ be the diagonal matrix whose entries are the correlation spectrum obtained as the solution of the CCA optimization problem. The **CCA norm** for a view in its canonical basis $\bar{\mathbf{z}}^{(\nu)} = W_\nu^\top \mathbf{z}^{(\nu)}$ is defined as:

$$\|\bar{\mathbf{z}}^{(\nu)}\|_{\text{CCA}}^2 = \sum_j^d \frac{1 - \lambda_j}{\lambda_j} \bar{z}_j^{(\nu)2} = \bar{\mathbf{z}}^{(\nu)\top} (I - \Lambda) \Lambda^{-1} \bar{\mathbf{z}}^{(\nu)}. \quad (4)$$

From (4) we see that dimensions where the between-view correlation is small have a large contribution to the CCA norm. Therefore, penalising with respect to the canonical norm induces a bias towards features that are highly correlated across the views and reduces the variance contributed by the uncorrelated features. In fact, under a weak assumption on the views, canonical ridge regression has been shown to have the following error with respect to the best predictor in either view (Kakade and Foster, 2007, Thm 1)

$$\mathbb{E}_{\mathcal{D}}[\operatorname{loss}(f_{\hat{\beta}}^{(\nu)})] - \operatorname{loss}(f^*) \leq \underbrace{5\varepsilon}_{\text{bias}} + \underbrace{\frac{1}{n} \sum_{j=1}^d \lambda_j^2}_{\text{variance}}. \quad (5)$$

The first term is the bias incurred by using the CCA norm regulariser whereas the second term bounds the variance of the prediction. Crucially, the *multi-view assumption* Kakade and Foster (2007) states that if a good predictor can be learned using each of the views individually, ε will be small. If the canonical correlation coefficients decay rapidly, since each $\lambda_j < 1$, the small increase in bias is made up for by the potentially large decrease in variance.

Generating Multiple Views. Although multiple views might exist in several problem domains, for many applications only a single view of the data is available. Furthermore, for the purposes of semi-supervised learning, obtaining two views which are equally good for prediction is crucial to achieving good performance with canonical ridge regression. If the multi-view assumption is violated, the representation learned using CCA will be poor resulting in a large bias term, ε . Correlated Nyström views (XNV) McWilliams et al. (2013) constructs two views from a single set of features by transforming the data using two independent sets of random feature functions guaranteeing that good predictors can be learned in both views.

However, the random feature transformations used by XNV are static. In this work we propose an adaptive multi-view approach where the views are given by neural networks whose weights are trained with respect to both the unsupervised CCA objective *and* the supervised loss. We detail this in the next section.

3. Canonically Correlated Networks

In this section we propose Canonically Correlated Networks (CCN) a deep multi-view architecture where the views are representations learned using neural networks. These networks are trained by backpropagation using both the CCA objective with a large amount of unlabeled data *and* a supervised loss on labeled data. This means that the views are not static as is the case with the shallow XNV model, but are adapted to both objectives. We first briefly review recent deep approaches CCA before describing the model architecture and training procedure which incorporates both unsupervised and supervised objectives as well as the CCA regularization.

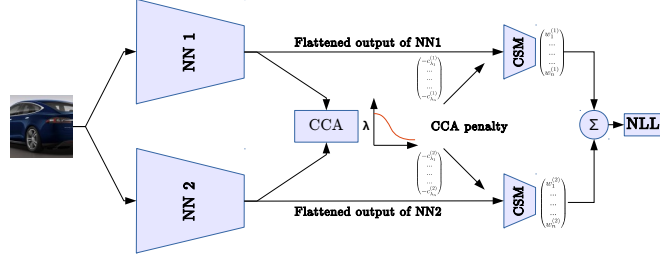


Figure 1: The CCN structure and the Canonical Softmax Layer in detail. The correlation spectrum obtained from the CCA pre-training allows us to compute the CCA norm. The weights of the CSM layer are penalized with the derivative of the CCA norm.

3.1 Deep Canonical Correlation Analysis

Recently, deep approaches to non-linear CCA have been proposed (Andrew et al., 2013; Wang et al., 2016), which learn non-linear, correlation maximizing transformations of each view as a set of parameters of an autoencoder model. Rather than finding linear basis vectors, deep CCA (DCCA) defines an optimization problem over the parameters of two neural networks.

With some abuse of notation, let $\mathbf{w} \in \mathbb{R}^d = [\text{vec}(W_1)^\top, \dots, \text{vec}(W_L)^\top]^\top$ now be the set of (vectorised) neural network weights which parameterize the non-linear transformation $g(\mathbf{x}; \mathbf{w})$ on input data, $\mathbf{x}$. We define view ν as the output of a network, $\mathbf{z}^{(\nu)} := g^{(\nu)}(\mathbf{x}; \mathbf{w}^{(\nu)})$. The notation $g^{(\nu)}$ indicates that the network architecture in each view need not be identical. Given two networks, the DCCA objective is to find the set of parameters for both networks jointly such that $\text{corr}(g^{(1)}(\mathbf{x}; \mathbf{w}^{(1)}), g^{(2)}(\mathbf{x}; \mathbf{w}^{(2)}))$ is maximized.

The correlation function does not decompose with respect to the samples – instead it is defined over the whole sample size. In order to have an objective which is tractable to evaluate and optimize for large datasets, it is necessary to approximate the correlation with respect to mini-batches of manageable size. For a batch size of b , the batch outputs are denoted in matrix form as $Z_\nu \in \mathbb{R}^{b \times d}$ whose rows are $\bar{\mathbf{z}}_i^{(\nu)\top}$, the centered views corresponding to a single mini-batch. We define the regularized empirical covariances of the centered views as

$$\hat{\Sigma}_{11} = \frac{1}{b} Z_1^\top Z_1 + r_1 I, \quad \hat{\Sigma}_{22} = \frac{1}{b} Z_2^\top Z_2 + r_2 I, \quad \hat{\Sigma}_{12} = \frac{1}{b} Z_1^\top Z_2,$$

where $r_1, r_2 > 0$ are regularisation parameters. These are necessary for the covariance matrices to be well conditioned since the batch size may be much smaller than the dimensionality of each view. To further improve the conditioning we accumulate statistics between minibatches. Forming the matrix $T = \hat{\Sigma}_{11}^{-1/2} \hat{\Sigma}_{12} \hat{\Sigma}_{22}^{-1/2}$, the deep CCA objective (the total correlation between views) is equivalent to

$$J_{\text{DCCA}}(\mathbf{w}^{(1)}, \mathbf{w}^{(2)}) = \text{corr}(Z_1, Z_2) = \text{trace}(\hat{\Sigma}_{11}^{-1/2} \hat{\Sigma}_{12} \hat{\Sigma}_{11}^{-1/2}). \quad (6)$$

Maximizing the DCCA objective requires computing gradients of this quantity with respect to the network parameters of each view (Andrew et al., 2013). This stochastic, minibatch optimization of the CCA criterion has been shown to achieve a good approximation of the true objective (Wang et al., 2016).

3.2 CCN model architecture and training

The model structure can be seen in figure 1. It consists of two parallel paths which each include a neural network which has been randomly initialised so that the generated views are different. In principal two

networks with different architectures could also be used (see §6). Here we describe the model in more detail with respect to the different training modalities.

Unsupervised training. In the unsupervised portion, the outputs of each of the views are used as input to the total correlation objective (6). In each mini-batch, we propagate the same input $\mathbf{x} \in \mathcal{D}_U$ through each of the neural networks in parallel. The two views are then centered to obtain $\bar{\mathbf{z}}^{(1)}, \bar{\mathbf{z}}^{(2)}$ and fed into the CCA layer, which implements the total correlation objective, Eq (6).

The correlation objective gradients with respect to the network parameters are computed by first taking the SVD of (6) then differentiating with respect to the network parameters in each view, analogously to Andrew et al. (2013). The weights are updated with respect to the total correlation by backpropagation using stochastic gradient descent with Nesterov momentum. The unsupervised portion of the training continues until J_{DCCA} saturates. The resulting model parameters are then used as the initialization for supervised training.

Canonical softmax. The *canonical softmax* (CSM) layer illustrated in figure 1 evaluates the negative log likelihood of the predicted classes and incorporates the canonical regularizer. In each mini-batch, we propagate input $\mathbf{x} \in \mathcal{D}_L$ through each of the neural networks as in the unsupervised mode. The CSM layer takes as input the corresponding flattened view which is in the deep CCA “basis” and the correlation spectrum, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$.

Unlike XNV which picks one of the views for the supervised learning step arbitrarily (eq. 3), we predict with and optimize the weights of the output layer corresponding to both views. The full supervised objective function containing an additional ℓ_2 -norm penalty is defined as follows:

$$J_{\text{CSM}} = - \sum_i^K \log p(y_i = l | x_i) + \frac{C}{2} (\|\mathbf{w}^{(1)}\|^2 + \|\mathbf{w}^{(2)}\|^2) + R_{\text{CCA}}^{(1)} + R_{\text{CCA}}^{(2)}.$$

$R_{\text{CCA}}^{(\nu)}$ is the canonical regularizer which penalizes the CCA norm of the weights in the CSM layer corresponding to view ν . Letting $\mathcal{W}_{\text{csm}}$ be the set of weights belonging to the CSM layer

$$R_{\text{CCA}}^{(\nu)} = \begin{cases} \frac{\gamma}{2} \cdot \|\mathbf{w}_S^{(\nu)}\|_{\text{CCA}}^2 & \text{if } S \in \mathcal{W}_{\text{csm}} \\ 0 & \text{else.} \end{cases} \quad (7)$$

Features are downweighted appropriately without any additional parameter tuning according to the decay of Λ but allowing $\gamma \in [0, 1)$ allows more fine-grained control over the overall CCA regularization strength. The ℓ_2 norm penalty is necessary since the CCA norm only penalises weights in the CSM layer.

We obtain the class activations, o_c as the sum of activations from the two views and obtain the class probabilities p_c using the softmax function $o_c = -(\mathbf{w}_c^{(1)\top} \bar{\mathbf{z}}^{(1)} + \mathbf{w}_c^{(2)\top} \bar{\mathbf{z}}^{(2)})$, $p_c = \text{softmax}(o_c)$, where $\mathbf{w}_c^{(\nu)}$ indexes the elements of the weight vector corresponding to class c . The gradients of the CSM layer are then computed with respect to the weights of each view independently to obtain

$$\frac{\partial J_{\text{CSM}}}{\partial \mathbf{w}_c^{(\nu)}} = \frac{\partial J_{\text{CSM}}}{\partial p_c} \cdot \frac{\partial p_c}{\partial o_c} \cdot \frac{\partial o_c}{\partial \mathbf{w}_c^{(\nu)}} + \frac{\partial R_{\text{CCA}}^{(\nu)}}{\partial \mathbf{w}_c^{(\nu)}} = (\delta_{kc} - p_c) \bar{\mathbf{z}}^{(\nu)} + C \mathbf{w}_c^{(\nu)} + \gamma (I - \Lambda) \Lambda^{-1} \mathbf{w}_c^{(\nu)},$$

where k is the index of the current label.

3.3 Projection, whitening and regularization

The CCA regularizer works by first projecting the input into the deep CCA basis. Recently other projection-based regularizers for neural networks have been proposed. Whitened neural networks (WNN) Desjardins et al. (2015) project the network parameters onto the manifold defined by the Fisher information matrix (FIM). In this space, dimensions are approximately uncorrelated. They maintain a set of weights in each layer, l : $W_l = U_l V_l$. U_l is a whitening matrix which is learned by factorizing the FIM rather than by minimizing the target loss and V_l is a matrix of model parameters learned via error backpropagation in the whitened space ($U_l \mathbf{x}_l$).

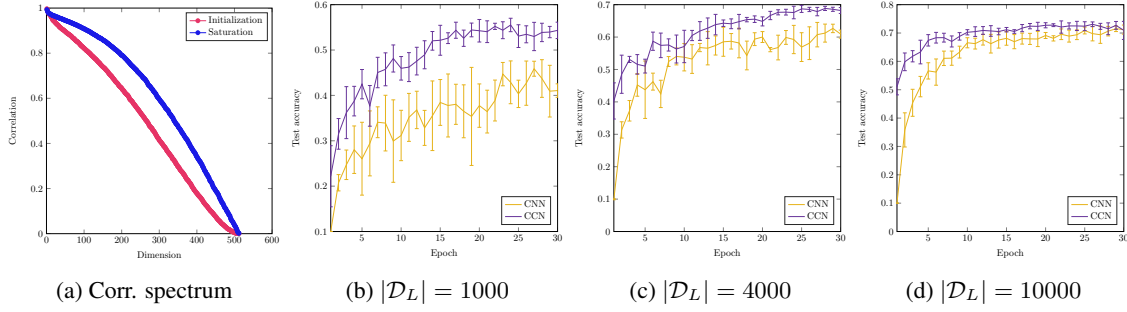


Figure 3: CIFAR-10: Correlation obtained by CCA and test accuracy for different sizes of $\mathcal{D}_L$.

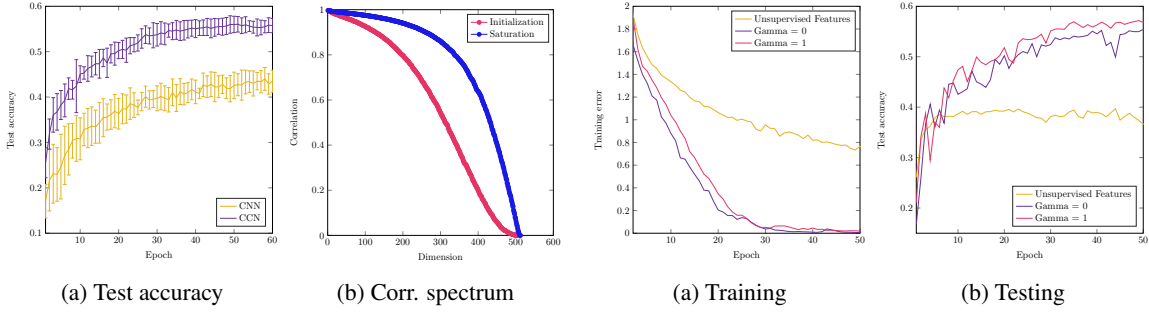


Figure 4: STL-10: Comparison of (a) test accuracy, (b) correlation spectrum before and after training.

Figure 5: Effect of canonical norm regularization on (a) training and (b) testing.

CCN on the other hand maintains only one set of model parameters $\mathbf{w}^{(\nu)}$. After optimizing the unsupervised objective, Eq (6), we see this simultaneously decorrelates features within each view and maximizes correlations them between views. This can be viewed as a type of regularized view whitening which improves the conditioning of the supervised optimization problem. We can view CCN as performing a projection into a (approximate) whitened space analogously to WNN: $(U^{(\nu)}\mathbf{z}^{(\nu)}) = \hat{\Sigma}_{\nu\nu}W^{(\nu)}\Lambda^{1/2}\mathbf{z}^{(\nu)}$ but where the projection and network parameters are coupled and learned by backpropagation. The whitening effect of CCA on the covariance matrix of activations in a single view can be seen in figure 2.

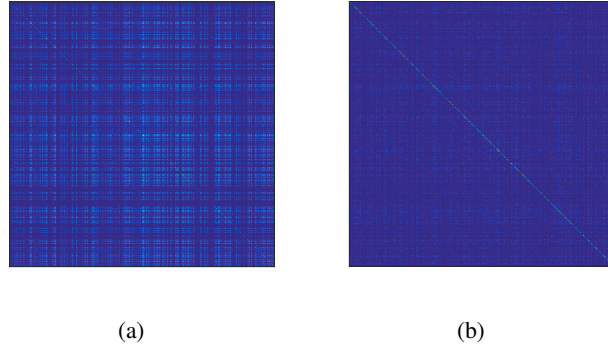


Figure 2: Comparison of the covariance matrix of activations (a) without CCA pre-training, and (b) after CCA.

4. Experiments

To evaluate the performance of our model in a semi-supervised setting, we performed experiments with convolutional networks on CIFAR-10, and STL-10. For each dataset we compare CCN against a baseline CNN. The baseline CNNs were chosen to match common architectures in the literature and the corresponding CCN model was chosen so that each view is roughly half the size of the baseline. In this way, CCN has the

same number of parameters or fewer when compared to the baseline. To control the capacity of the baseline, we used dropout regularization as well as batch normalization.

Implementation details. CCN training was performed using stochastic gradient descent with Nesterov momentum. Unsupervised training was performed until the CCA objective saturated, typically between 1-3 epochs. All model hyper-parameters were tuned using a held-out validation set. We used spatial batch normalization (Ioffe and Szegedy, 2015) in all layers.

The main bottleneck of CCN is constructing and inverting the correlation matrix, T . When the output dimensionality of the views is large, this is computationally expensive. To tackle this problem we employ the randomized SVD (Halko et al., 2009). This returns a randomized rank r approximation to the full singular value decomposition setting all $r + 1, \dots, d$ singular values (corresponding to canonical correlation coefficients) to be zero, effectively performing dimensionality reduction.

The canonical regularizer imposes a severe penalty on dimensions with low correlation coefficients. This leads to large gradient values which deteriorate the performance significantly. To avoid this phenomenon, we also use a cutoff threshold, τ , for the canonical softmax gradient calculation. Weights corresponding to dimensions for which $\lambda_j < \tau$ are set to zero, effectively sparsifying the output.

CIFAR-10. We use a model consisting of three convolutional layers with filter sizes 32, 32 and 64 and using 3×3 average pooling at the output of every layer followed by a fully connected layer with 512 units. The CCN

Table 1: CIFAR: Test accuracy mean and variance.

$ \mathcal{D}_L =$	1,000	4,000	10,000
CNN	0.4278(± 0.0352)	0.5996(± 0.0313)	0.7022(± 0.0186)
CCN	0.5395(± 0.0199)	0.6825(± 0.0101)	0.7225(± 0.0160)

model uses this topology in both views which are subsequently concatenated and fed through the CSM layer. We use rectified linear units throughout as nonlinearities. The baseline model is similar in topology but features a larger number of convolutional filters per layer: 64, 128 and 256 respectively and the fully-connected layer features 512 units. This is similar to the model used by Paine et al. (2014).

We compare CCN against the baseline CNN with labeled subsets of size $|\mathcal{D}_L| = \{1, 000, 4, 000, 10, 000\}$. $\mathcal{D}_U$ consists of a 300,000 image subset sampled from the Tiny Images dataset. Images were normalized using global contrast normalization and ZCA whitening (Coates et al., 2011) and no data augmentation was used. Results on the test set are shown in table 1. We observe that CCN reaches a higher mean accuracy with lower variance for a smaller number of training epochs. The relative performance gap decreases as more labeled samples are used for training.

STL-10. STL consists of 5,000 labeled and 100,000 unlabeled 96×96 images. We used the standard ten-fold procedure described in (Coates et al., 2011) so $|\mathcal{U}_L| = 1,000$ per fold. Images were preprocessed using global contrast normalization and no data augmentation was used. We use a deeper model to control the dimensionality at the output layer of each view. The model consist of five convolutional layers normalized using spatial batch normalization. The respective sizes of the layers are 32, 32, 64, 64 and 128 filters per layer using a filter size of 5×5 in the lower three layers and 3×3 in the upper two layers. The baseline model topology is once again similar, but contains 64, 64, 128, 128 and 256 filters per layer respectively.

Averaged over the ten folds the CNN baseline achieved a mean test accuracy of $0.4338(\pm 0.0238)$ and CCN achieved an accuracy of $0.5552(\pm 0.0155)$ (fig. 4a). The stark improvement of CCN over the baseline is due to the fast decay of the correlation spectrum (fig. 4b) which imposes stronger regularization on the uncorrelated features leading to substantially reduced variance. We also ran the experiments with a standard data augmentation pipeline, resulting in a mean test accuracy of $0.5472 (\pm 0.0192)$ for the baseline ² and $0.6112 (\pm 0.0163)$ for CCN.

Effect of canonical norm regularization. Figure 5 shows the effect of the canonical norm regularizer on a single fold of STL-10. We compare the performance of CCN when we set $\gamma = \{0, 1\}$ (c.f. equation 7)

2. The baseline CNN achieves accuracy comparable to the baseline presented in Zhao et al. (2015).

and the situation when after unsupervised training, the convolutional layers are fixed (i.e. only the weights of the softmax layer are updated during supervised training and no canonical norm penalty is used). The latter compares to the “supervised” setting presented in (Wang et al., 2016). We observe that adapting the convolutional layers during supervised training greatly improves over simply using CCA to generate features. Adding the canonical norm penalty further improves the training and test performance.

5. Related work

A common semi-supervised approach within the context of deep learning involves unsupervised pretraining of the models in a greedy layer-wise fashion and subsequently fine-tuning the weights using a supervised objective. Most notably, Restricted Boltzmann Machine pretraining and different variations of the stacked autoencoder paradigm (Bengio et al., 2007; Jarrett et al., 2009; Erhan et al., 2010). Generative modelling is another common approach to semi-supervised learning. Recent examples include variational autoencoders (Kingma et al., 2014), Stacked What-Where autoencoders (Zhao et al., 2015) and generative adversarial networks (Springenberg (2015)).

Most similarly to CCN, Weston et al. (2012) suggest employing an auxiliary nonlinear embedding objective which is applied to one or more layers in the network. The network is then trained simultaneously on both objectives. Of particular recent interest are *ladder networks* (Rasmus et al., 2015), which present a novel way of performing simultaneous training on both supervised and unsupervised objectives using a denoising objective at every layer.

6. Discussion

We have presented CCN, a deep architecture for semi-supervised learning motivated by theoretical approaches to multi-view learning. We have shown that, similar to the shallow case, regularizing a supervised objective with the canonical norm of deep representations also obtains estimators with lower variance – and ultimately better generalisation performance – compared with a baseline convolutional network when the number of labeled samples is small.

Unlike e.g. dropout, CCA regularization appears to perform complementary regularization to batch normalization, showing significant improvements *especially* when used in conjunction with batch normalization – attributable to the proper scaling of the CCA gradients throughout the network. In fact, dropout has been shown to act as an ℓ_2 regularizer which is adaptive to a *diagonal* approximation to the FIM (Wager et al. (2013) (c.f. §3.3)). Since only the diagonal is used, it cannot adapt to correlations between features. This may explain why dropout has been observed to perform similar regularization to batch normalization (Ioffe and Szegedy (2015) (a phenomenon we also observe empirically)).

The problem of regularization in recurrent neural networks is a less studied and more difficult problem (Krueger and Memisevic (2016); Cooijmans et al. (2016)). We have attempted to instantiate the CCN model with one layer LSTM sequence-to-one models on text classification datasets but found that despite an initial boost for the first few epochs of training, the baseline LSTM network converges to a better solution. This is due to the lack of consensus in how to apply batch normalization – which CCN is reliant on – in recurrent

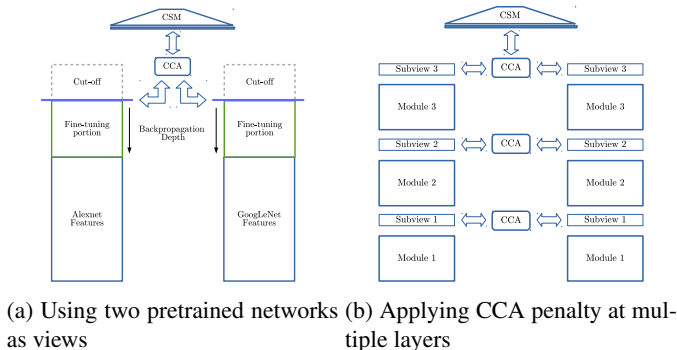


Figure 6: Alternative CCN architectures.

networks, thus making it difficult to decouple the effect of canonical regularization from other factors. We aim to investigate this in further work.

The main drawback of CCN is that in general deep CCA presents a difficult optimization problem, especially in large scale settings. It requires performing matrix inversions and computing SVDs of high-dimensional matrices which is computationally expensive although recent approaches have sought to make the CCA objective more amenable to stochastic optimization (Wang et al., 2016).

Overall, the encouraging performance boost afforded by CCA regularisation leaves many open avenues for future research. Obvious extensions include multi-view domain adaptation (Blitzer et al., 2011) and investigating more general measures of dependence. We elaborate on two particularly interesting directions.

Leveraging pre-trained networks. Many large state-of-the-art networks (e.g. AlexNet, GoogLeNet (Szegedy et al., 2015), VGG (Simonyan and Zisserman, 2014)) are available as pre-trained reference models. One possible extension is to use these pre-trained models as initialisations for the two views which can then be fine-tuned by the CCA and regularized supervised learning step (figure 6a). **CCA-as-a-layer.** CCA and the associated canonical regularizer can be viewed as a more general “layer” (figure 6b). Choosing an appropriate correlation threshold can impose a form of sparsity on the weights – setting those which have low correlation between views to zero. Inserting a CCA layer at several points would have the effect of sparsifying the network further and would also decouple the projection operation and weight matrices between layers (c.f. discussion in §3.3).

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